

Data Analysis and Storytelling

Regression analysis - hierarchical models II

Lionel Hertzog

Geodata Paris - IGN
lionel.hertzog@ign.fr

FIRS - Track 1

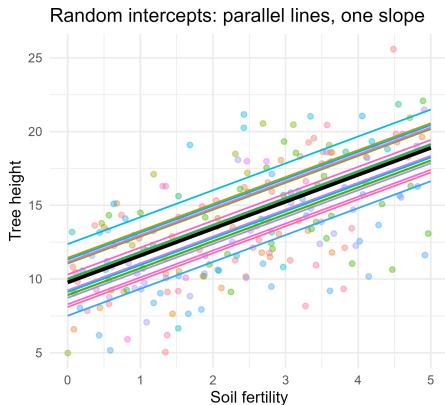
From flexible intercepts to flexible slopes

We are interested to understand how soil fertility influence tree height. We measured trees and soil fertility across a set of plot. A corresponding hierarchical model can be:

$$height_{ij} \sim N(\mu_{ij}, \sigma_{res})$$

$$\mu_{ij} = \beta_0 + \beta_1 * fertility_i + \gamma_j$$

$$\gamma_j \sim N(0, \sigma_{plot})$$



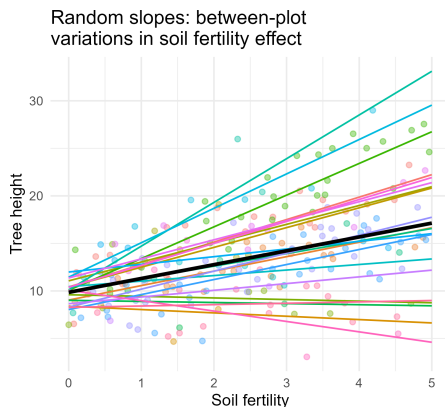
From flexible intercepts to flexible slopes

Any regression coefficients can be made flexible in a hierarchical structure, including slopes. For instance, the slope for the effect of soil fertility can be made to vary between plots in a hierarchical setting:

$$height_{ij} \sim N(\mu_{ij}, \sigma_{res})$$

$$\mu_{ij} = \beta_0 + (\beta_1 + \delta_j) * fertility_i + \gamma_j$$

$$\delta_j \sim N(0, \sigma_{slope}), \gamma_j \sim N(0, \sigma_{int})$$



About varying slopes

In essence varying slopes are similar to interaction, from our example a non-hierarchical model would be:

$$height_{ij} \sim N(\mu_{ij}, \sigma_{res})$$

$$\mu_{ij} = \beta_j + \beta_1 * fertility + \beta_{2j} * fertility$$

What are the limits of such a model?

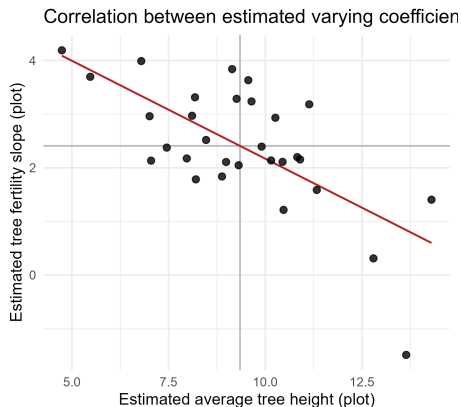
About varying slopes

The limits of interaction terms are the same as the no pooling models we saw in the previous unit:

- Inefficient when the number of level becomes large (> 10)
- Does not use information across levels
- Does not allow for prediction to new level

Covariation between hierarchical terms

- The varying coefficients can be unrelated as seen in the previous slides.
- It is also possible to estimate correlation between the varying coefficients.
- From our tree height example, negative correlation means that plots with higher tree height tend to have more negative relation to soil fertility



Covariation between hierarchical terms

The model can be written followingly:

$$height_{ij} \sim N(\mu_{ij}, \sigma_{res})$$

$$\mu_{ij} = \beta_0 + \gamma_j + (\beta_1 + \delta_j) * fertility_i$$

$$\begin{pmatrix} \gamma_j \\ \delta_j \end{pmatrix} \sim MVN\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_{int}, \rho \\ \rho, \sigma_{slope} \end{pmatrix}\right)$$

With *MVN* a multivariate normal distribution and ρ the correlation between the varying intercepts and the varying coefficient for the effect of fertility.

Selection of hierarchical terms

To build the hierarchical structure several heuristics can be used:

- Automatic inclusion of varying intercepts to account for "design" variables, such as plots
- Selection of covariable with varying slopes based on: (i) a priori research question (i.e. what is the amount of variability in soil fertility effect on tree height?), (ii) varying slopes added during model building to fit model assumptions (i.e. residual checks showing patterns), (iii) selection based on statistical tests.
- Correlation between hierarchical terms, by default off, added if needed based on research question or to fit model assumptions.